Four ways AI is making the power grid faster and more resilient

A review of the article by June Kim in Nov 2023 presented by Bryce King

A Complicated Future

- The grid faces challenges from a number of different quarters adding to the complexity of the grid
  - Increasing Demand
  - Distributed Energy
  - Intermittent Renewables
  - Extreme Weather
- Governments and utilities are looking to AI for possible solutions

AI Capabilities

- AI can reason in real time over complex scenarios and provide humans with insights that can improve the grid in major ways
- Article outlines four major impacts
  - 1. Faster and better decision-making
  - 2. Tailored approach for every home
  - 3. Making EVs work with the grid
  - 4. Spotting disasters before they hit

Faster & Better Decision Making

Faster & Better Decision Making

- The power grid is too large and complex for humans to process all at once
- AI can help operators
  - Understand current conditions
  - Make better decisions
  - Predict potential problems

In the Real World

- Midcontinent Independent System Operator (MISO) has tested machine-learning models that optimize daily planning
- Run calculations to predict next day electricity demand and determine cost-effective dispatch plans
- AI helps run these calculations 12x faster
- Future efforts to predict outage risks to prevent outages and minimize damages

Tailored Approaches for Every Home

Tailored Approaches for Every Home

- Data about energy usage can help consumers save money and grid operators plan more efficient, more reliable supply
- Smart meters and the internet of things give consumers the power to be more energy efficient
- Smart technology can interact with the grid and let consumers know when to use energy intensive appliances to help smooth out demand

In the Real World

- Lunar Energy, a batter and grid-technology startup helps customers optimize energy usage
- Gridshare helps individuals choose when to charge electric vehicles
- Consumers can choose to provide utility companies with charging information to help them plan energy dispatch

Making EVs work for the grid

Making EVs work for the grid

- EVs add enormous energy demand that complicates grid function
- Demand is usually clustered around certain cities and neighborhoods
- Predicting and smoothing demand for EV charging can help balance the grid
- EVs can be used as distributed energy storage, providing energy back to the grid when demand is high

In the Real World

- WeaveGrid collaborates with utility companies, automakers, and charging companies to analyze EV charging data
- DTE Energy (Michigan) has partnered with WeaverGrid to improve grid planning
- Customers can receive texts recommending optimal charging times

Spotting Disasters Before They Hit

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Spotting Disasters Before They Hit

- Disasters can be caused by faulty equipment, bad infrastructure, downed trees, and severe weather, along with other things
- AI can be used to inspect equipment and identify damage or overgrown trees at risk of causing damage
- AI forecasting can help grid operators prepare for bad weather, predict high demand, and anticipate renewable energy output

In the Real World

- Rhizome, a startup in Washington, is using AI to predict grid failures from extreme weather by training on utility historical data and energy equipment performance
- Keen AI uses helicopters and drones to identify rust on transmission towers
- Nvidia is working on better weather modeling and simulation

What’s Next?

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- Likely can’t take humans out of the loop
  - AI can enhance mathematical calculations, but the risks involved in grid management require incredibly reliability that AI has yet to prove
  - Data privacy poses an issue with smart metering and EV tracking
  - AI models can perpetuate bias by continuing to underserve historically underserved neighborhoods
- More and more technology will include AI in the loop to make recommendations, make predictions, and automate decisions

Thank you